

Exam Date & Time: 25-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Social Network Analysis [CSE-4074]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast adjacency matrices and edge lists as methods for representing network data. In what scenarios is one representation preferable over the other? (5)

A)

- B) Compute the degree of each node and derive the network using the degree distribution of the following graph.

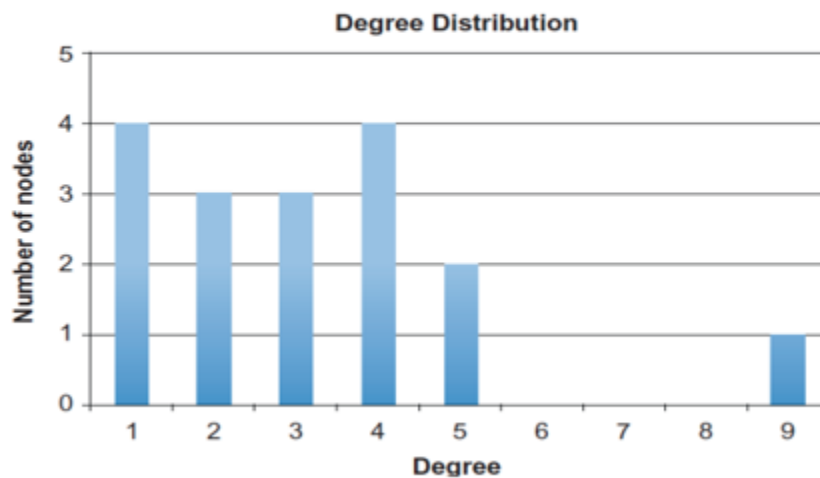


Figure 1: The Graph for reasoning

- 2) (5)

A)

Find the local clustering coefficient with respect to Node D and E.

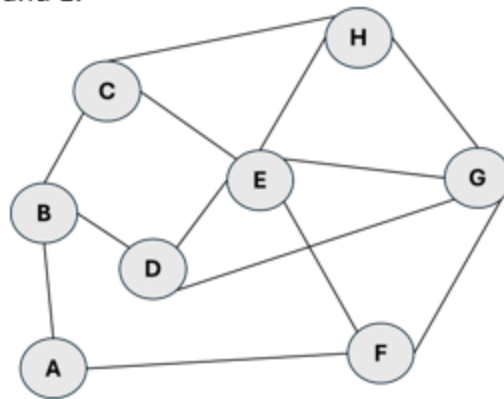


Figure 3: The Graph for reasoning

- B) Compare and contrast Degree centrality, and Betweenness centrality with respect to the following diagram. Identify the highest centrality node in terms of Degree centrality, and Betweenness centrality.

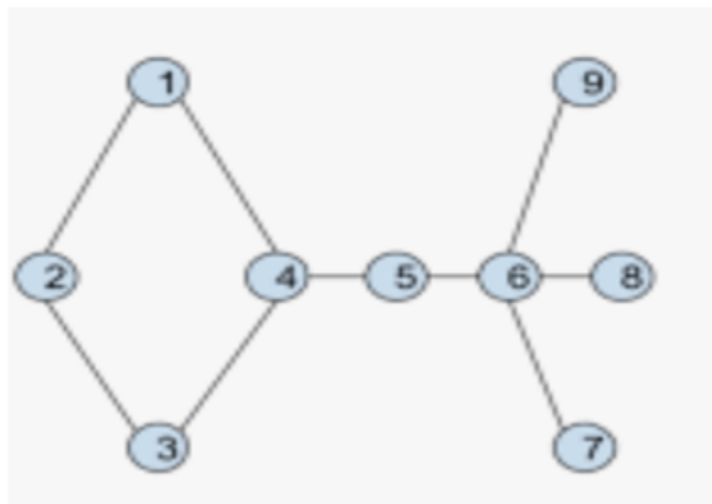


Figure 4: The Graph for reasoning

- 3) Explain the relationship between tie strength and network structure. Outline and give examples of the two key principles of tie strength. (5)
- A)
- B) Examine the concepts of strong ties and weak ties in social network analysis. How do these types of connections differ in terms of interaction frequency, trust, and information flow? Provide examples of scenarios where strong ties are beneficial and others where weak ties offer unique advantages within a network. (5)
- 4) (5)
- A)

Compute Laplacian Matrix and Normalized Laplacian matrix for this graph.

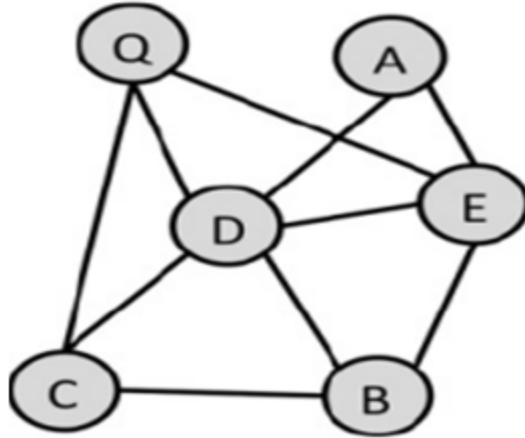


Figure 5: The Graph for reasoning

- B) You are analysing a large online social network with thousands of users, and your goal is to identify distinct communities based on user interactions. After applying the normalized cut method, you observe that the network has been partitioned into several meaningful groups. Examine the significance of the cut value in determining the quality of the community partitions. (5)
- 5) Compare and contrast the significance of influence-related statistics in social network analysis. (5)
- A)
- B) Describe the concept of homophily in social network analysis. How does homophily influence the formation of connections within a network, and what are some examples of attributes that drive homophily in social networks? (5)

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